جامعة الطائف
TAIF UNIVERSITY



WIRELESS SENSOR AIR QUALITY MONITORING SYSTEM

Cleaner air, better life

Project course:
Computer networks

Supervised by Professor:
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PROJECT IDEA :

The Smart Air Quality Monitoring System Using Wireless Sensors represents an advanced Smart City solution designed to improve environmental monitoring through the integration of wireless sensing technologies, intelligent networking infrastructure, and real-time data analysis. The proposed system deploys multiple wireless sensor nodes across different geographical locations to continuously measure critical environmental parameters, including PM2.5, PM10, carbon dioxide (CO_2), volatile organic compounds (VOC), temperature, and humidity.

The collected information is transmitted through a wireless communication network to a centralized monitoring platform, where the data is processed, analyzed, and visualized in real time. In addition, the system incorporates an automated alert mechanism capable of notifying both authorities and end users whenever pollutant concentrations exceed predefined safety thresholds. By providing continuous environmental awareness, the proposed solution contributes to healthier urban environments and supports evidence-based decision-making processes.

NETWORK TYPE :

The proposed project falls under the category of Smart City Networks, as it enables distributed environmental monitoring across multiple urban locations while maintaining centralized supervision and control. Such systems are considered a fundamental component of modern smart city infrastructures that aim to improve sustainability, public health, and quality of life.

GENERAL GOAL :

The primary goal of this project is to establish a reliable and scalable network infrastructure capable of supporting real-time environmental monitoring and facilitating the efficient collection, transmission, and analysis of air quality data across urban environments.

PROBLEM STATEMENT :

Rapid industrialization and urban growth have significantly contributed to increasing levels of air pollution worldwide. Although conventional monitoring stations provide valuable environmental information, their deployment is often limited by high installation costs, maintenance requirements, and restricted geographical coverage.

As a result, environmental authorities frequently encounter difficulties in obtaining accurate and real-time information regarding pollution distribution and severity. Delays in detecting hazardous environmental conditions may reduce the effectiveness of preventive measures and negatively impact public health.

Therefore, there is a growing need for an intelligent, scalable, and cost-effective monitoring system capable of providing continuous environmental visibility while supporting timely interventions and informed decision-making.

WHY DO WE NEED THIS SOLUTION?

The proposed solution addresses the limitations of traditional monitoring approaches by offering:

- Continuous real-time monitoring.
- Wider geographical coverage.
- Reduced deployment and operational costs.
- Immediate alert generation.
- Enhanced environmental awareness.
- Improved support for environmental planning and policy-making.

OBJECTIVES :

The proposed system contributes to the development of sustainable smart cities by providing an intelligent and scalable environmental monitoring solution. It enables real-time air quality monitoring, supports informed decision-making, and promotes public health and environmental sustainability.

SPECIFIC OBJECTIVES INCLUDE:

- Monitoring air quality indicators continuously across multiple locations.
- Enhancing environmental sustainability through data-driven analysis.
- Providing early warning notifications when pollution levels exceed safe limits.
- Improving communication reliability and network performance.
- Strengthening network security through logical segmentation techniques.
- Facilitating centralized monitoring and management.
- Supporting future expansion and integration with additional IoT devices and smart city services.

NETWORK TOPOLOGY :

The network adopts a Star Topology architecture in which all sensor nodes, monitoring devices, and servers communicate through a centralized switching infrastructure.

JUSTIFICATION FOR SELECTING STAR TOPOLOGY

The Star Topology was selected due to its numerous operational advantages, including:

- Simplified network administration and troubleshooting.
- High reliability and fault isolation capabilities.
- Ease of scalability and future expansion.
- Improved network performance.
- Compatibility with modern IoT-based architectures.
- Reduced impact of individual device failures on overall network functionality.

VLAN SEGMENTATION:

Future Network Enhancement

To further improve network performance and security, VLAN technology can be incorporated in future implementations. VLANs would allow logical separation between sensors, monitoring devices, and administrative systems, reducing unnecessary traffic and improving network management.

VLAN BENEFITS:

Implementing VLANs in this architecture provides critical network optimization through two main mechanisms:

- Broadcast Domain Reduction: Continuous data streaming from numerous air quality sensors can generate high broadcast traffic, leading to network congestion. By isolating sensors into their own VLAN, broadcast traffic is contained, preventing network degradation and ensuring the central server and monitoring workstations maintain optimal performance.
- Enhanced Security and Logical Isolation**:** Air quality sensors deployed in public urban areas are vulnerable to physical or remote tampering. VLAN segmentation ensures that if a sensor node is compromised, the attacker is logically contained within the sensor network and cannot directly access the central server, financial logs, or administrative workstations.

IMPLEMENTATION: NETWORK SIMULATION

The proposed network is implemented using Cisco Packet Tracer as the primary simulation and validation platform

NETWORK COMPONENTS:

- Router
- Managed Switch
- Central Server
- Monitoring Workstations (PCs)
- Wireless Sensor Nodes
- Wireless Communication Infrastructure

IMPLEMENTATION TASKS:

The implementation process includes:

- Network topology design.
- Device configuration.
- Wireless sensor integration.
- Server deployment.
- Environmental data monitoring.
- Connectivity verification.

The network was designed to support communication between IoT sensors and monitoring devices through a centralized infrastructure. Environmental data is transmitted wirelessly through the Access Point and made available to monitoring stations for observation and analysis.

IMPLEMENTATION: HARDWARE PROTOTYPE

To validate the system architecture beyond network simulation, a physical prototype of the Wireless Sensor Node was successfully developed and tested. This hardware integration bridges the gap between physical environmental data acquisition and network communication.

HARDWARE COMPONENTS:

- **Microcontroller**: ESP32 NodeMCU development board equipped with an integrated Wi-Fi module for processing and wireless communication.
- **Air Quality Sensor:** MQ Gas Sensor used to continuously monitor and measure critical environmental parameters.
- **Local Visualizer:** 0.96" OLED I2C Display for immediate, real-time visualization of the monitored data.
- **Infrastructure:** Solderless breadboard and jumper wires for structural circuit layout and secure electrical connectivity.

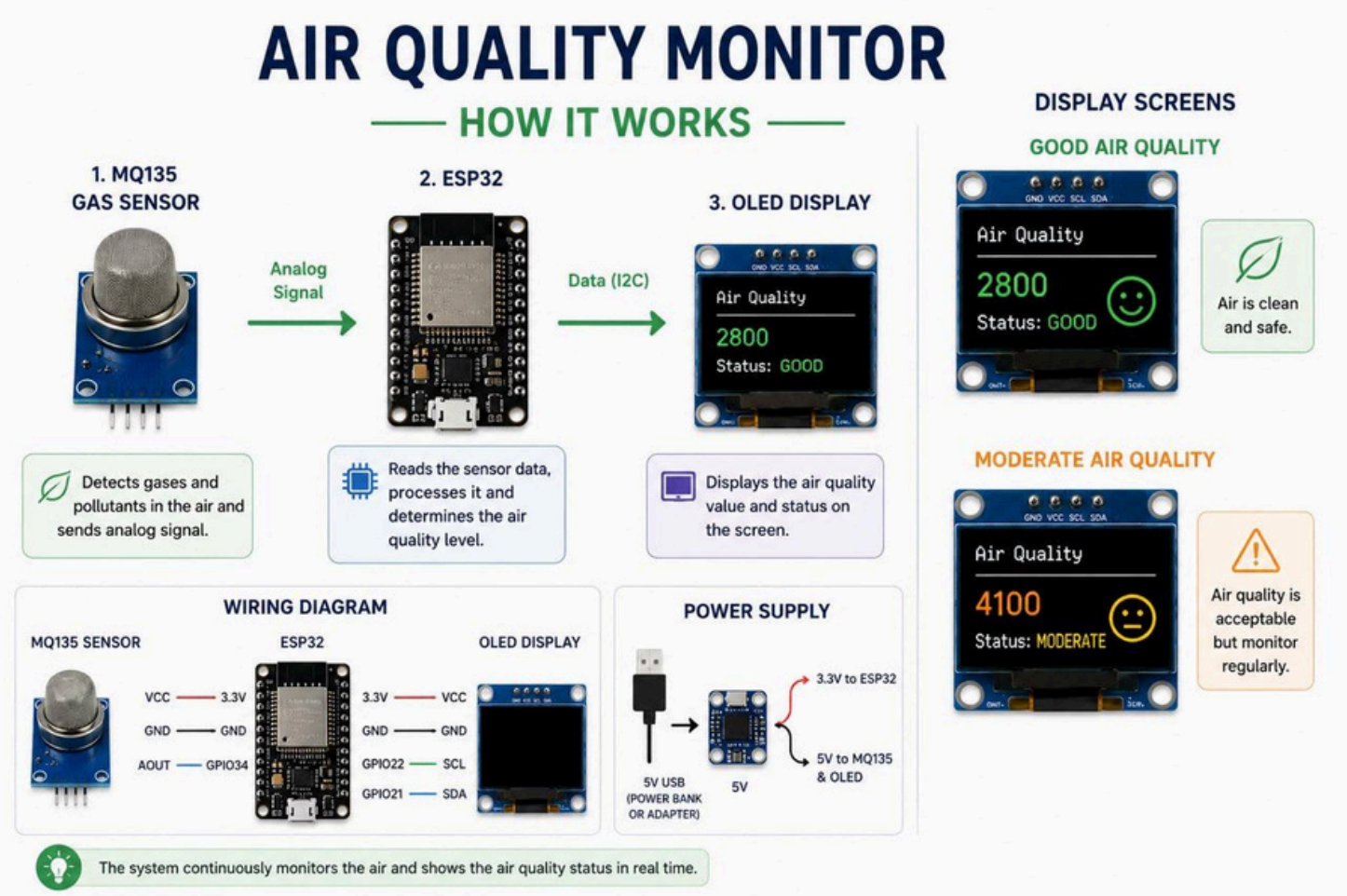


Figure 1 – Separated Hardware Components of the Sensor Node

This figure presents the hardware components and operational workflow of the wireless sensor node. It illustrates how the MQ135 gas sensor sends data to the ESP32 microcontroller for edge processing, which then transmits the metrics via I2C to be visualized on the OLED display in real time.

WIRING AND OPERATION WORKFLOW:

- **Data Acquisition:** The MQ sensor is wired directly to the input pins of the ESP32 to feed real-time atmospheric data into the processor.
- **Edge Display**: The ESP32 utilizes the I2C protocol to render air quality metrics locally on the OLED screen for continuous visualization.
- **Wireless Network Connection:** Utilizing its built-in Wi-Fi interface, the ESP32 node establishes a wireless connection to the local network architecture. This enables the physical device to function as an active edge node, capable of transmitting environmental packet data across the Star Topology to a central monitoring infrastructure.

VIDEO DEMONSTRATION:

This QR code links to a video demonstration showing the real-time operational testing of the physical hardware node, verifying its data acquisition and edge-display performance.

Click here to watch the video

RESULTS :

Following the implementation phase, the network demonstrated successful operation and achieved all intended design objectives.

Key results include:

- Successful communication between all network devices.
- Reliable transmission of environmental data from sensor nodes to the central server.
- Effective wireless communication through the Access Point.
- Efficient centralized monitoring using the server and monitoring devices.
- Support for real-time environmental monitoring.
- Scalable architecture suitable for future expansion.

The simulation demonstrated successful operation of the proposed system and confirmed the ability of the network to support environmental monitoring applications.

ANALYSIS :

The obtained results indicate that the proposed network architecture successfully fulfills the requirements of a modern Smart City environmental monitoring system.

The Star Topology provided a highly reliable and manageable network structure while simplifying maintenance and fault detection procedures. Furthermore, the wireless communication infrastructure enabled efficient interaction between IoT sensors and monitoring devices, supporting continuous environmental monitoring.

The successful simulation results demonstrate the practicality and effectiveness of the proposed design. Moreover, the network exhibits strong scalability characteristics, allowing additional sensors and monitoring devices to be integrated without significantly affecting system performance.

Overall, the proposed architecture offers an efficient and future-ready solution suitable for large-scale smart city deployments.

IMAGES AND COMMENTS :

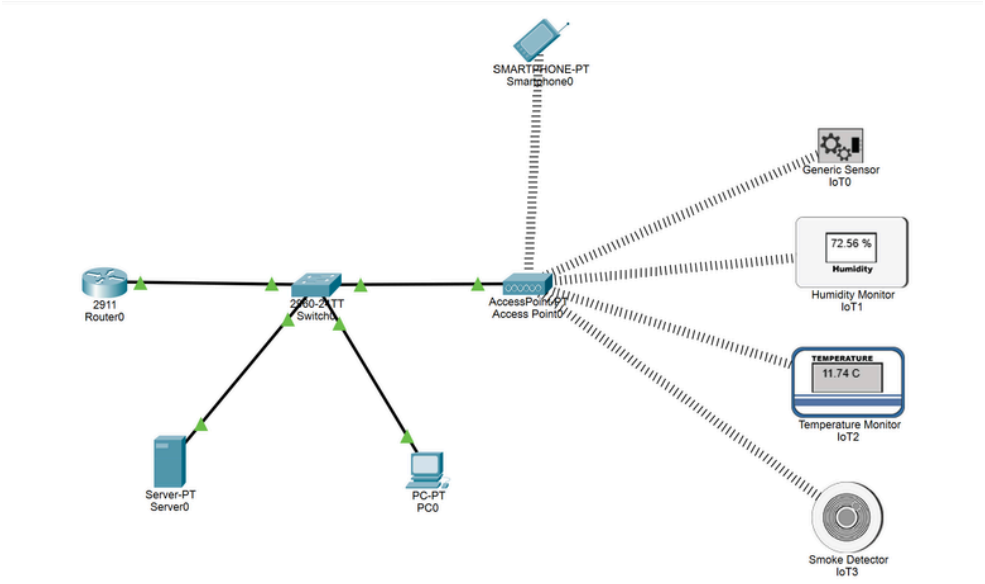


Figure 2 – Smart Air Quality Monitoring System Topology

This figure presents the overall network architecture of the proposed Smart Air Quality Monitoring System. The design integrates IoT environmental sensors, including temperature, humidity, smoke detection, and air quality monitoring devices, with a centralized networking infrastructure composed of a router, switch, access point, server, and monitoring devices. Operating under a Star Topology configuration, the system enables reliable wireless communication, real-time environmental monitoring, and efficient data management, making it suitable for modern Smart City applications.

IMAGES AND COMMENTS :

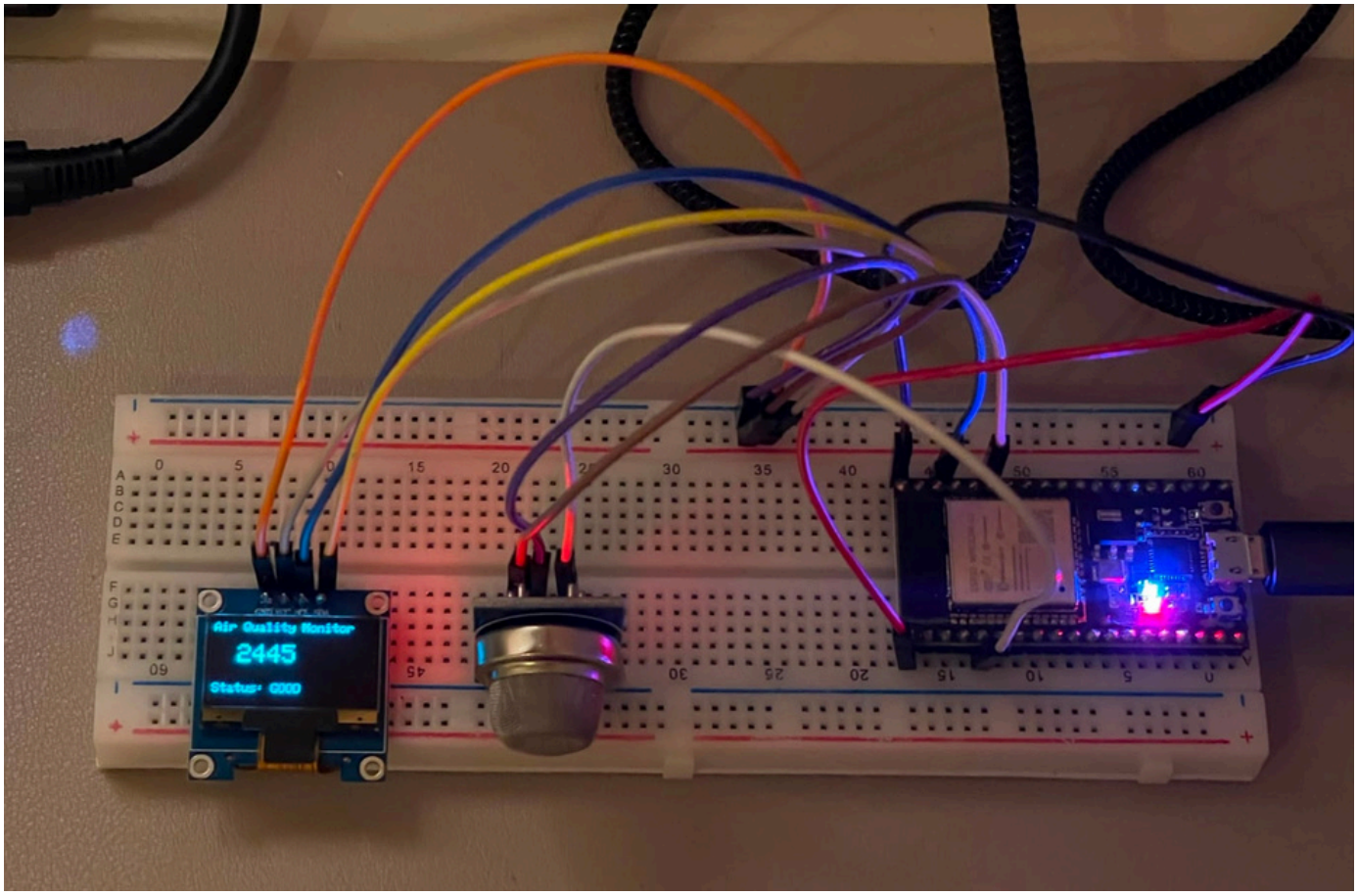


Figure 3 - Smart Air Quality Monitoring System Hardware

This figure presents the overall hardware architecture of the proposed Smart Air Quality Monitoring System. The design integrates an ESP32 microcontroller with an MQ air quality sensor and an OLED display. Operating under a Star Topology configuration, the system enables reliable wireless communication, real-time environmental monitoring, and efficient data management, making it suitable for modern Smart City applications.

CONCLUSION :

This project presented the design and implementation of a Smart Air Quality Monitoring System Using Wireless Sensors within a Smart City environment. By integrating wireless sensing technologies with modern networking principles, the proposed solution enables real-time environmental monitoring, improves data accessibility, and supports proactive environmental management.

The implementation results confirmed the effectiveness of the proposed architecture in achieving reliable communication, efficient data management, and effective environmental monitoring. The adoption of Star Topology and wireless communication technologies enhanced network scalability, maintainability, and operational efficiency, making the system a promising solution for future smart city applications.